



**CSE-476: Introduction to Natural Language Processing**  
**Quiz 3**

**Multiple Choice Questions (Single correct)**

*Instructions: Select the correct option with a short explanation.*

- Q1. In sequence tasks, what output summarizes the entire processed input?
- A. To store the final output of the entire sequence.
  - B. To act as a memory that carries information from previous time steps
  - C. To calculate the loss function for the current word
  - D. To normalize the input embeddings
- Q2. Which option specifically aims to avoid very large gradients during training?
- A. To ensure the output values sum to 1, representing a probability distribution
  - B. To prevent the gradients from exploding
  - C. To increase the speed of the training process
  - D. To allow the model to handle variable-length sequences
- Q3. Which option incorrectly claims that neural networks cannot use vector embeddings?
- A. They cannot use embeddings
  - B. They assume fixed input size and struggle with temporal order
  - C. They cannot be trained with gradient descent
  - D. They do not use activation functions
- Q4. During optimization, which step is responsible for updating the network weights?
- A. To update weights based on the loss
  - B. To calculate the predicted output and compute the loss
  - C. To unroll the network into a sequence
  - D. To calculate the gradient for each parameter

- Q5. In an unstable training setup, which option describes weights that change randomly every step?
- A. They are randomly re-initialized at every step
  - B. They are reused across timestamps
  - C. They increase in value as the sequence progresses
  - D. Only the output weight  $V$  is reused
- Q6. Which option claims that using more layers always yields a deeper model with
- A. A deeper network with more representational power
  - B. A single linear model
  - C. A recurrent neural network
  - D. A decision tree
- Q7. In free-running generation (no teacher forcing), the model predicts the next word using what history?
- A. The next word given the predicted history
  - B. The next word given the ground-truth history
  - C. The current word given itself
  - D. All words simultaneously
- Q8. In a simplified model that ignores context, the hidden state is updated from what at time  $t$ ?
- A. Only the input at time  $t$
  - B. Only the hidden state at time  $t - 1$
  - C. Both the input at time  $t$  and the hidden state at time  $t - 1$
  - D. All future inputs
- Q9. If  $W$  is chosen so that its rows correspond to input features and
- A.  $(n_0 \times n_1)$
  - B.  $(n_1 \times n_0)$
  - C.  $(n_0 \times 1)$
  - D.  $(1 \times n_1)$
- Q10. During deployment, which option best describes the step where the model generates outputs?
- A. To compute predictions during inference
  - B. To calculate gradients of the loss with respect to parameters
  - C. To initialize weights randomly
  - D. To normalize input data